PROJECT PRESENTATION

CivicResolve

AI-Based Public Grievance Management System

Smart Complaints. Clear Decisions. Better Governance.



Presented by: [Your Name] | [Department / College]

Problem Statement

Manual effort

Officials may spend time checking complaints and forwarding them to the correct department.

Inconsistent decisions

Priority and escalation can vary when clear criteria are not applied consistently.

Limited transparency

Citizens may struggle to understand complaint progress or why a decision was made.

The need

A structured system that validates complaint data, recommends actions using explicit rules, explains its reasoning, and keeps officials responsible for final decisions.

Proposed Solution & Objectives

CivicResolve combines a digital workflow with symbolic AI.

Validate structured complaint fields using CSP constraints

Apply Forward Chaining IF–THEN rules

Recommend department, priority, and escalation

Explain recommendations using the actual reasoning trace

Support tracking, assignment, updates, and closure

Symbolic AI

Uses predefined facts, constraints, and production rules—not ML or an LLM.

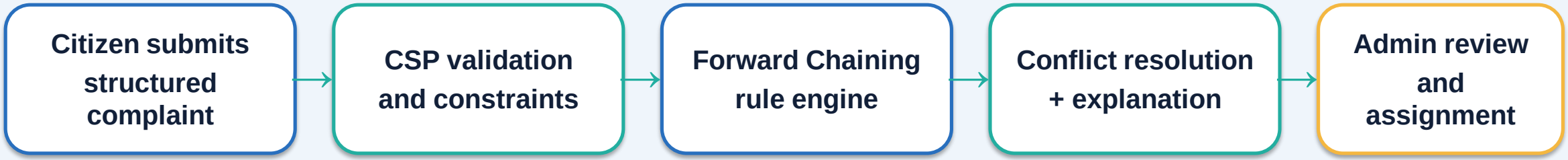
Human in control

Administrators review recommendations and retain authority over final actions.

Core goal

Consistent, traceable grievance handling.

System Architecture



Operational workflow



Unresolved validation conflicts are flagged for correction or authorized review.

Symbolic AI: CSP + Forward Chaining

1. CSP validation

• Variables: category, location, severity, urgency

• Domains: permitted values

• Constraints: required fields and consistency conditions

Invalid cases are flagged; user facts are not silently changed.

2. Forward Chaining

• Start with validated facts

• Evaluate IF–THEN rules

• Add derived facts to working memory

• Repeat until no new facts can be derived

3. Conflict + explanation

• Priority precedence: CRITICAL > HIGH > MEDIUM > LOW

• Preserve rules and facts in the trace

• Generate a template-based explanation

• Refer unresolved conflicts for review

Example: validated high-severity and urgency facts may activate a predefined HIGH-priority rule.

Modules & User Roles

Citizen

Register complaints • View status • Track history • Receive supported updates

Administrator

Review recommendations • Assign cases • Record overrides • Verify closure

Department officer

View assigned complaints • Process cases • Post updates • Record resolution

Core system modules

01 Authentication & role management

02 Complaint registration & CSP validation

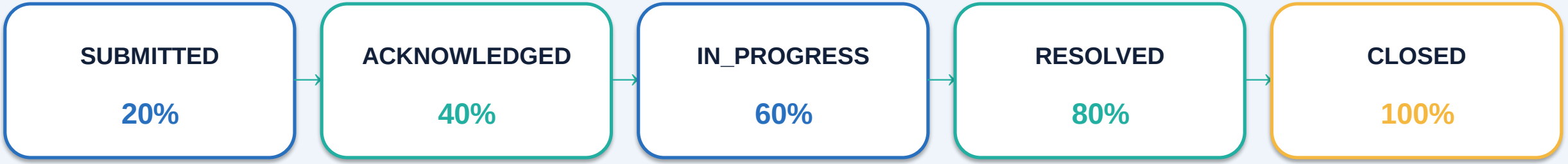
03 Knowledge base & inference engine

04 Conflict resolution & explanation

05 Assignment & officer management

06 Tracking, notifications & dashboard

Complaint Lifecycle



Responsibilities

Administrator acknowledges, assigns, reviews, and closes.

Officer processes the assigned complaint and records updates.

Progress percentages represent workflow stages. They do not guarantee the percentage of physical work completed.

Implementation & Testing Plan

Implementation components

- Web interface for the three roles
- Complaint data store and status history
- CSP validation rules
- Production-rule knowledge base
- Forward Chaining and conflict handling
- Explanation and audit trace

Key test cases

- Valid and invalid complaint inputs
- High / critical priority rule activation
- Multiple priority recommendations
- Explanation matches fired rules
- Role permissions and assignment
- Invalid status transition and closure checks

Report only the test results actually observed; do not claim accuracy or success rates without measurements.

Research Context & Contribution

Related research

- Digital grievance and citizen participation
- Online service quality
- Explainable AI in government
- Human–AI decision-making

CivicResolve contribution

- CSP validation of structured fields
- Forward Chaining recommendations
- Explicit priority precedence
- Deterministic explanations
- Human-supervised decisions

Limitations

- Depends on predefined rules
- Cannot verify real-world truth by itself
- Unseen cases may need review
- Effectiveness must be measured through testing

Design principle: recommendations should be explainable, reviewable, and accountable.

Conclusion & Future Work

Conclusion

CivicResolve combines structured grievance management with symbolic AI to validate complaint data, derive transparent recommendations, and support consistent tracking. Human officials retain responsibility for final decisions.

Future enhancements

- Expand and version the knowledge base
- Improve conflict handling and audit reports
- Add multilingual and accessibility support
- Strengthen privacy and security
- Evaluate with larger real-world test sets

Thank you | Questions?